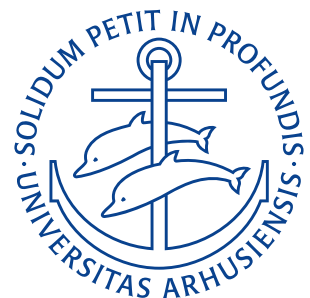


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Exercises

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Lecture 1: Fourier Theory

29. August 2025

Problem 1.1

Consider the following periodic functions. Find possible values for the period, using the definition of periodic functions.

- a) $f(x) = e^{\cos x}$
- b) $f(x) = \cos\left(x + \frac{\pi}{3}\right)$
- c) $f(x) = \cos\left(\frac{\pi x}{3}\right)$
- d) $f(x) = \cos x + \cos(3x)$
- e) $f(x) = \cos(x) \cos(3x)$
- f) $f(x) = \cos x + \cos(0,6x)$

Derivation

- a) From the lecture (p. 1), we know that a function f is periodic, with period p , if

$$f(x) = f(x + p).$$

Using this definition we obtain:

$$\begin{aligned} e^{\cos x} &= e^{\cos(x+p)} \\ \cos x &= \cos(x + p) \\ \Rightarrow p &= 2\pi, 4\pi, 6\pi, \dots \end{aligned}$$

- b) We use the same definition of periodicity to obtain

$$\begin{aligned} \cos\left(x + \frac{\pi}{3}\right) &= \cos\left(x + p + \frac{\pi}{3}\right) \\ \Rightarrow p &= 2\pi, 4\pi, 6\pi. \end{aligned}$$

- c) Following the same procedure we get

$$\begin{aligned} \cos\left(\frac{\pi x}{3}\right) &= \cos\left(\frac{\pi(x+p)}{3}\right) \\ \cos\left(\frac{\pi x}{3}\right) &= \cos\left(\frac{\pi x}{3} + \frac{\pi p}{3}\right) \\ \Rightarrow \frac{\pi p}{3} &= 2\pi, 4\pi, 6\pi, \dots \\ \Rightarrow p &= 6, 12, 18, \dots \end{aligned}$$

- d) We follow the same procedure and get

$$\begin{aligned} \cos x + \cos(3x) &= \cos(x + p) + \cos(3(x + p)) \\ \cos x + \cos(3x) &= \cos(x + p) + \cos(3x + 3p). \end{aligned}$$

This holds if p is an integer multiple of 2π and $3p$ is an integer multiple of 2π , i.e. $p = 2\pi, 4\pi, 6\pi, \dots$

- e) Once again, we follow the procedure and obtain

$$\begin{aligned} \cos(x) \cos(3x) &= \cos(x + p) \cos(3(x + p)) \\ \cos(x) \cos(3x) &= \cos(x + p) \cos(3x + 3p). \end{aligned}$$

From the above, we see that $p = \pi, 2\pi, 3\pi$. This is due to the fact, that the two functions are multiplied, hence $\cos(0) \cdot \cos(0) = \cos(\pi) \cos(3\pi) = 1$

f) We follow the same procedure as in the previous subproblems and obtain

$$\cos x + \cos(0,6x) = \cos(x + p) + \cos(0,6x + 0,6p).$$

I.e. p must be an integer multiple of 2π and $0,6p$ must be an integer multiple of 2π . Hence $p = 10\pi, 20\pi, 30\pi, \dots$

- a) $p = 2\pi, 4\pi, 6\pi, \dots$
- b) $p = 2\pi, 4\pi, 6\pi, \dots$
- c) $p = 6, 12, 18, \dots$
- d) $p = 2\pi, 4\pi, 6\pi, \dots$
- e) $p = \pi, 2\pi, 3\pi, \dots$
- f) $p = 10\pi, 20\pi, 30\pi, \dots$

RESULT

Problem 1.2

Let

$$f(x) = x, \quad -1 \leq x \leq 1, \quad f(x) = f(x + p), \quad p = 2.$$

- Sketch f
- Find a Fourier series representation of f .

Derivation

- On [Figure 0.1](#) the periodic repeating function $f(x)$ is depicted.

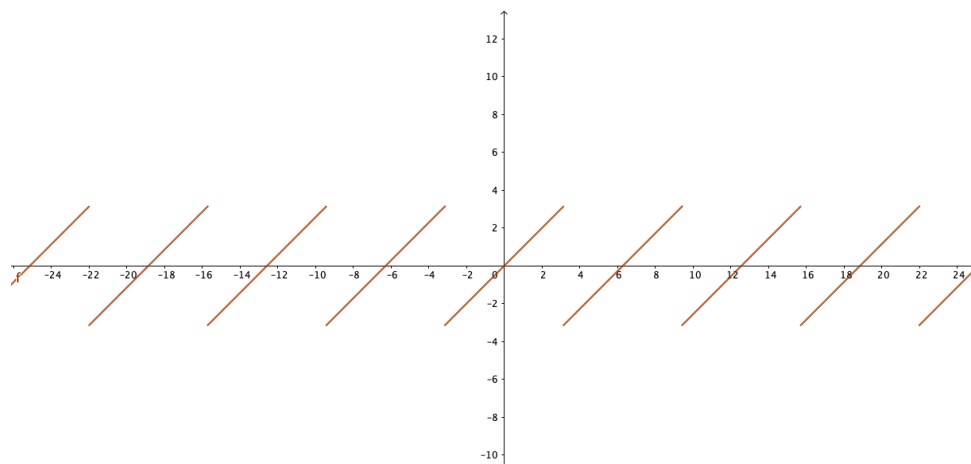


Figure 0.1: Sketch of f .

- From the lecture (p. 5), we have the general representation of a Fourier series as

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

for

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \quad n = 1, 2, 3, \dots$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx, \quad n = 1, 2, 3, \dots$$

In this case, $L = \frac{p}{2} \Rightarrow L = 1$. Therefore:

$$a_0 = \frac{1}{2} \int_{-1}^1 x dx$$

$$= \frac{1}{2} \left[\frac{x^2}{2} \right]_{-1}^1$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{2} \right)$$

$$= 0$$

$$a_n = \int_{-1}^1 x \cos(n\pi x) dx$$

$$\begin{aligned}
&= 0 \\
b_n &= \int_{-1}^1 x \sin(n\pi x) \, dx \\
&= \left[\frac{\sin(n\pi x)}{\pi^2 n^2} - \frac{x \cos(n\pi x)}{\pi n} \right]_{-1}^1 \\
&= \left[-\frac{x \cos(n\pi x)}{\pi n} \right]_{-1}^1 \\
&= \left(-\frac{\cos(n\pi)}{\pi n} + \left(\frac{\cos(-n\pi)}{\pi n} \right) \right) \\
&= \left(-\frac{\cos(n\pi)}{\pi n} - \frac{\cos(n\pi)}{\pi n} \right) \\
&= -\frac{2}{\pi n} \cos(n\pi).
\end{aligned}$$

Hence, the complete Fourier representation is

$$f(x) = \sum_{n=1}^{\infty} -\frac{2}{\pi n} \cos(n\pi) \sin(n\pi x).$$

a) See Figure 0.1

b)
$$f(x) = \sum_{n=1}^{\infty} -\frac{2}{\pi n} \cos(n\pi) \sin(n\pi x).$$

RESULT

Problem 1.3

Let $a \neq 0$ and $b \neq 0$. Let $f(x) = f(x + p_0)$ be a periodic function with period p_0 . Is $f(ax + b)$ periodic? If yes, find a period. Give arguments for your answer.

Derivation

We let $f(x) = f(x + p)$ denote the function given in the assignment and $g(x) = f(ax + b)$ be the function which we wish to determine is periodic or not. To determine if $g(x)$ is periodic we must determine if $g(x) = g(x + p)$ holds. We get:

$$\begin{aligned} g(x + p) &= g(x) \\ &= f(a(x + p) + b) \\ &= f(ax + b + ap). \end{aligned}$$

Comparing this with the original function f we see that it repeats when $ap = p_0$. Since $a \neq 0$ we can divide by a and obtain

$$p = \frac{p_0}{a}.$$

The function is periodic with period $p = \frac{p_0}{a}$

RESULT

Problem 1.4

Calculate the left-hand limit and the right hand limit of

$$f(x) = \frac{|x|}{x}$$

at $x = 0$.

Derivation

From the lecture (p. 2), we know that the left hand limit of $f(x)$ at x_0 is defined as

$$f(x_0-) = \lim_{h \rightarrow 0} f(x_0 - h)$$

and the right hand limit of $f(x)$ at x_0 is

$$f(x_0+) = \lim_{h \rightarrow 0} f(x_0 + h)$$

for positive h .

Using these two definitions we get:

$$\begin{aligned} f(0-) &= \lim_{h \rightarrow 0} f(0 - h) \\ &= \lim_{h \rightarrow 0} \frac{|-h|}{-h} \\ &= \lim_{h \rightarrow 0} \frac{|h|}{-h} \\ &= -1 \end{aligned}$$

$$\begin{aligned} f(0+) &= \lim_{h \rightarrow 0} f(0 + h) \\ &= \lim_{h \rightarrow 0} \frac{|h|}{h} \\ &= 1. \end{aligned}$$

$\begin{aligned} f(0-) &= -1 \\ f(0+) &= 1. \end{aligned}$
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RESULT

Problem 1.5

Calculate the left-hand derivative and the right-hand derivative of

$$f(x) = x|x|$$

at $x = 0$.

Derivation

From the lecture (p. 4) the definition of the left hand derivative of $f(x)$ at x_0 is defined as

$$f'(x_0-) = \lim_{h \rightarrow 0} \frac{f(x_0) - f(x_0 - h)}{h}$$

and the right hand derivative of $f(x)$ at x_0 is defined as

$$f'(x_0+) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

for positive h .

Using these definitions we get:

$$\begin{aligned} f'(0-) &= \lim_{h \rightarrow 0} \frac{f(0) - f(-h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-(-h|-h|)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h|h|}{h} \\ &= \lim_{h \rightarrow 0} |h| \\ &= 0 \end{aligned}$$

$$\begin{aligned} f'(0+) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h|h|}{h} \\ &= \lim_{h \rightarrow 0} |h| \\ &= 0. \end{aligned}$$

$\begin{aligned} f'(0-) &= 0 \\ f'(0+) &= 0. \end{aligned}$

RESULT

Lecture 2: Even and odd functions

5. September 2025

Problem 2.1

Find a period and the corresponding Fourier coefficients of

- a) $f(x) = \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{2x}{4}\right) + \cos\left(\frac{\pi}{5}\right) \sin\left(\frac{3x}{7}\right)$
- b) $f(x) = \sin(\sqrt{2}x) + \cos\left(\frac{x}{\sqrt{2}}\right)$
- c) $f(x) = \sin\left(\frac{\pi}{3} + \frac{3\pi x}{4}\right)$.

Hint: You may want to use

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \sin \beta \cos \alpha.$$

Derivation

- a) We start by rewriting the given function as

$$f(x) = \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{7\pi x}{14\pi}\right) + \cos\left(\frac{\pi}{5}\right) \sin\left(\frac{6\pi x}{14\pi}\right).$$

Now, comparing with the general form of a Fourier series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

We see that $L = 14\pi$, $p = 28\pi$, $b_6 = \cos(\pi/5)$ and $b_7 = \sin(\pi/4)$. All other Fourier coefficients are 0.

- b) We rewrite as:

$$f(x) = \sin\left(\frac{2\pi x}{\sqrt{2}\pi}\right) + \cos\left(\frac{\pi x}{\sqrt{2}\pi}\right).$$

By comparison with the general form we see that $L = \sqrt{2}\pi$, $p = 2\sqrt{2}\pi$, $b_2 = 1$, and $a_1 = 1$. All other Fourier coefficients are 0.

- c) We use the hint to rewrite as

$$\begin{aligned} f(x) &= \sin\left(\frac{\pi}{3} + \frac{3\pi x}{4}\right) \\ &= \sin\left(\frac{\pi}{3}\right) \cos\left(\frac{3\pi x}{4}\right) + \cos\left(\frac{\pi}{3}\right) \sin\left(\frac{3\pi x}{4}\right). \end{aligned}$$

By comparison, we see that $L = 4$, $p = 8$, $a_3 = \sin(\pi/3)$, and $b_3 = \cos(\pi/3)$. All other Fourier coefficients are zero

a)	$p = 28\pi,$	$b_6 = \cos\left(\frac{\pi}{5}\right),$	$b_7 = \sin\left(\frac{\pi}{4}\right),$
b)	$p = 2\sqrt{2}\pi,$	$a_1 = 1,$	$b_2 = 1,$
c)	$p = 8,$	$a_3 = \sin\left(\frac{\pi}{3}\right),$	$b_3 = \cos\left(\frac{\pi}{3}\right).$

RESULT

Problem 2.2

Are the following functions even or odd? Give arguments.

a) $f(x) = (\cos(x^2) + \cos(x)) \sin(x)$

b) $f(x) = \frac{1 + \cos x}{2 + \cos(x^2)}$

c) $f(x) = \frac{1 + \sin(x)}{2 + \sin(x^2)}$

d) $f(x) = x|x|$.

Derivation

From the lecture (p. 3), we know that a function f is even if

$$f(x) = f(-x) \quad \text{for all } x$$

and that a function f is odd if

$$f(x) = -f(-x) \quad \text{for all } x.$$

We will therefore try to insert $-x$ in the place of x and see if that causes a change of sign for each of the given functions.

We will however, instead utilize the simple fact that multiplication of functions follow the same rules for even/odd as multiplication of numbers. Furthermore, we know that x is odd, $\cos$ is even, $\sin$ is odd and $|x|$ is even. In the following, odd functions are called O and even functions are called E .

a) Using the method described above we get:

$$\begin{aligned} f(x) &= (\cos(x \cdot x) + \cos(x)) \sin(x) \\ &= \cos(x \cdot x) \sin x + \cos(x) \sin(x) \\ &= E(O \cdot O) \cdot O + E \cdot O \\ &= E \cdot O + E \cdot O \\ &= O + O \\ &= O. \end{aligned}$$

b) Again,

$$\begin{aligned} f(x) &= \frac{1 + \cos x}{2 + \cos(x \cdot x)} \\ &= \frac{E + E}{E + E} \\ &= \frac{E}{E} \\ &= E. \end{aligned}$$

c) Again,

$$\begin{aligned} f(x) &= \frac{1 + \sin x}{2 + \sin(x \cdot x)} \\ &= \frac{E + O}{E + O(E)} \\ &= \frac{E + O}{E + E} \end{aligned}$$

$$\begin{aligned} &= \frac{E + O}{E} \\ &= \frac{E}{E} + \frac{O}{E} \\ &= E + O. \end{aligned}$$

As the sum of an even and an odd function is neither even nor odd if none of the functions are 0. Hence the function in question is neither even nor odd.

d) Lastly,

$$\begin{aligned} f(x) &= x \cdot |x| \\ &= O \cdot E \\ &= O. \end{aligned}$$

- a) Odd,
- b) Even,
- c) Neither even nor odd,
- d) Odd.

RESULT

Problem 2.3

Consider the periodic function

$$f(x) = \begin{cases} x, & -2 < x < -1 \\ x + k, & -1 < x < 1 \\ x, & 1 < x < 2 \end{cases}$$

$$f(x) = f(x + p), p = 4$$

where k is a constant. Decompose $f(x)$ into its even and odd part.

Derivation

For $-2 < x < -1$ we get

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}(x + (-x)) = 0$$

$$f_2(x) = \frac{1}{2}(f(x) - f(-x)) = \frac{1}{2}(x - (-x)) = x.$$

For $-1 < x < 1$ we get

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}(x + k + (-x + k)) = k$$

$$f_2(x) = \frac{1}{2}(f(x) - f(-x)) = \frac{1}{2}(x + k - (-x + k)) = x.$$

And for $1 < x < 2$ we get

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}(x + (-x)) = 0$$

$$f_2(x) = \frac{1}{2}(f(x) - f(-x)) = \frac{1}{2}(x - (-x)) = x.$$

We thus have the decomposed function $f(x)$

$$f(x) = f_1(x) + f_2(x)$$

with

$$f_1(x) = \begin{cases} 0, & -2 < x < -1 \\ k, & -1 < x < 1 \\ 0, & 1 < x < 2. \end{cases}$$

$$f_1(x) = f_1(x + p), \quad p = 4$$

and

$$f_2(x) = x, \quad -2 < x < 2$$

$$f_2(x) = f_2(x + p), \quad p = 4.$$

$$\begin{aligned} f(x) &= f_1(x) + f_2(x) \\ f_1(x) &= \begin{cases} 0, & -2 < x < -1 \\ k, & -1 < x < 1 \\ 0, & 1 < x < 2. \end{cases} \\ f_1(x) &= f_1(x+p), \quad p=4 \\ f_2(x) &= x, \quad -2 < x < 2 \\ f_2(x) &= f_2(x+p), \quad p=4. \end{aligned}$$

RESULT

Problem 2.4

Consider the periodic function

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 2x, & 0 < x < \pi \end{cases}$$

$$f(x) = f(x + p), p = 2\pi.$$

Decompose $f(x)$ into its even and odd part and find its Fourier series.

Hint: You may want to use examples from the lecture.

Derivation

For $-\pi < x < 0$ we get

$$f_1(x) = \frac{1}{2} (f(x) + f(-x)) = \frac{1}{2} (0 + (-2x)) = -x$$

$$f_2(x) = \frac{1}{2} (f(x) - f(-x)) = \frac{1}{2} (0 - (-2x)) = x.$$

For $0 < x < \pi$ we get

$$f_1(x) = \frac{1}{2} (f(x) + f(-x)) = \frac{1}{2} (2x + (0)) = x$$

$$f_2(x) = \frac{1}{2} (f(x) - f(-x)) = \frac{1}{2} (2x - (0)) = x.$$

Therefore

$$f_1(x) = |x|, -\pi < x < \pi.$$

From the lecture (p. 7) the Fourier cosine series of this is

$$f_1(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} (\cos(n\pi) - 1) \cos(nx).$$

Also

$$f_2(x) = x, -\pi < x < \pi.$$

Which has the Fourier sine series (p. 6)

$$f_2(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n} \sin(nx).$$

Combining these two Fourier series we get the complete Fourier series:

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left(\frac{2}{\pi n^2} (\cos(n\pi) - 1) \cos(nx) + (-1)^{n+1} \frac{2}{n} \sin(nx) \right).$$

$$f_1(x) = |x|, \quad -\pi < x < \pi$$

$$f_2(x) = x, \quad -\pi < x < \pi$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left(\frac{2}{\pi n^2} (\cos(n\pi) - 1) \cos(nx) + (-1)^{n+1} \frac{2}{n} \sin(nx) \right).$$

RESULT

Problem 2.5

Decompose

$$f(x) = e^{ix}$$

into its even and odd part.

Hint; you may want to use the representation of $f(x)$ in terms of $\cos x$ and $\sin x$.

Derivation

We rewrite the function as

$$f(x) = e^{ix} = \cos x + i \sin x.$$

Now, to decompose we follow the same procedure as above:

$$\begin{aligned} f_1(x) &= \frac{1}{2} (f(x) + f(-x)) \\ &= \frac{1}{2} (\cos x + i \sin x + \cos(-x) + i \sin(-x)) \\ &= \frac{1}{2} (\cos x + i \sin x + \cos x - i \sin(x)) \\ &= \cos x \\ f_2(x) &= \frac{1}{2} (f(x) - f(-x)) \\ &= \frac{1}{2} (\cos x + i \sin x - \cos(-x) - i \sin(-x)) \\ &= \frac{1}{2} (\cos x + i \sin x - \cos x + i \sin x) \\ &= i \sin x. \end{aligned}$$

$\begin{aligned} f_1(x) &= \cos x \\ f_2(x) &= i \sin x. \end{aligned}$

RESULT